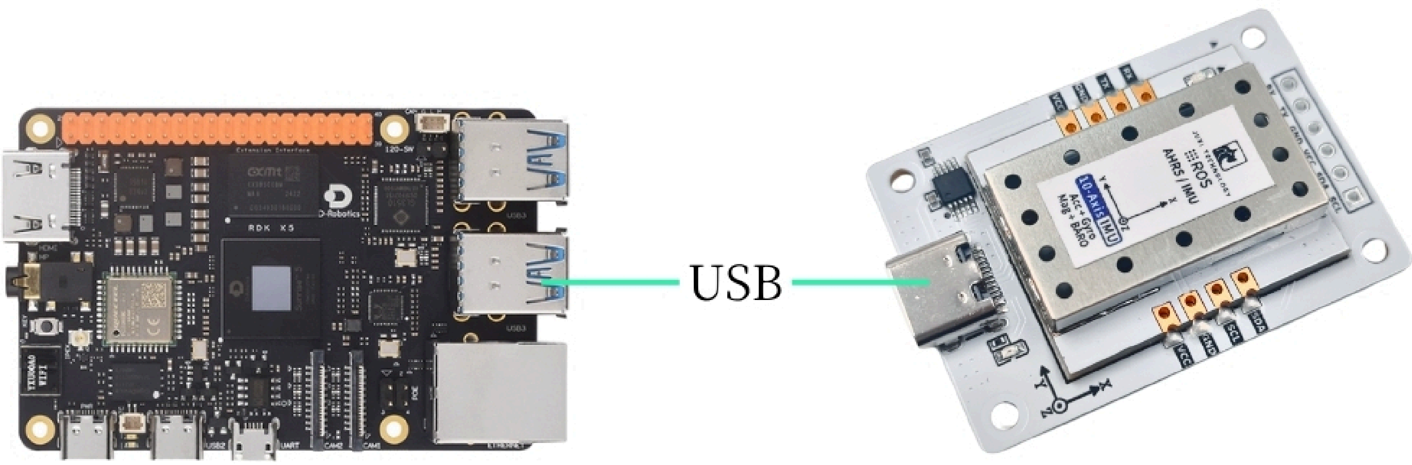
RDK Series

1. Connect the device

This tutorial takes the mirroring of the? version of the RDK X5 motherboard as an example.

Connect the IMU attitude sensor to the USB port of the main controller via a Type-C cable.



2. Check device status

View device ID

代码块

```
lsusb
```

View Device Number

代码块

```
ls -l /dev/ttyU*
```

```
[redacted]@ubuntu:~$ lsusb
Bus 002 Device 002: ID 05e3:0626 Genesys Logic, Inc. USB3.1 Hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 003: ID 1a86:7523 QinHeng Electronics CH340 serial converter
Bus 001 Device 002: ID 05e3:0610 Genesys Logic, Inc. Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
[redacted]@ubuntu:~$
```

```
[redacted]@ubuntu:~$ ll /dev/ttyU*
crw-rw---- 1 root dialout 188, 0 Oct 21 17:57 /dev/ttyUSB0
[redacted]@ubuntu:~$
```

Set Port Mapping

代码块

```
# 防止插拔后端口变更，请设置端口映射
sudo gedit /etc/udev/rules.d/99-serial-imu.rules

# 如出现没有gedit命令相关内容，先下载安装
sudo apt install gedit

# 填写映射内容
KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523", MODE:="0777", SYMLINK+="imu-serial"

# 参数说明：
`--mode`: 通信模式，可选值为`serial`(串口)或`i2c`
`--port`: 串口名(如`/dev/ttyUSB0`)或I2C端口号(如`7`)
`--rate`: 数据打印频率(Hz)，默认10Hz
`--debug`: 启用调试模式，显示详细信息

# 保存退出，运行命令使规则生效
sudo udevadm trigger
sudo service udev reload
sudo service udev restart

# 验证
ll /dev/imu-serial

# 输出示例：
lrwxrwxrwx 1 root root 7 1月 22 10:00 /dev/imu-serial -> ttyUSB0
```

3. Install the driver library

3.1 Install the Python libraries required for the code

代码块

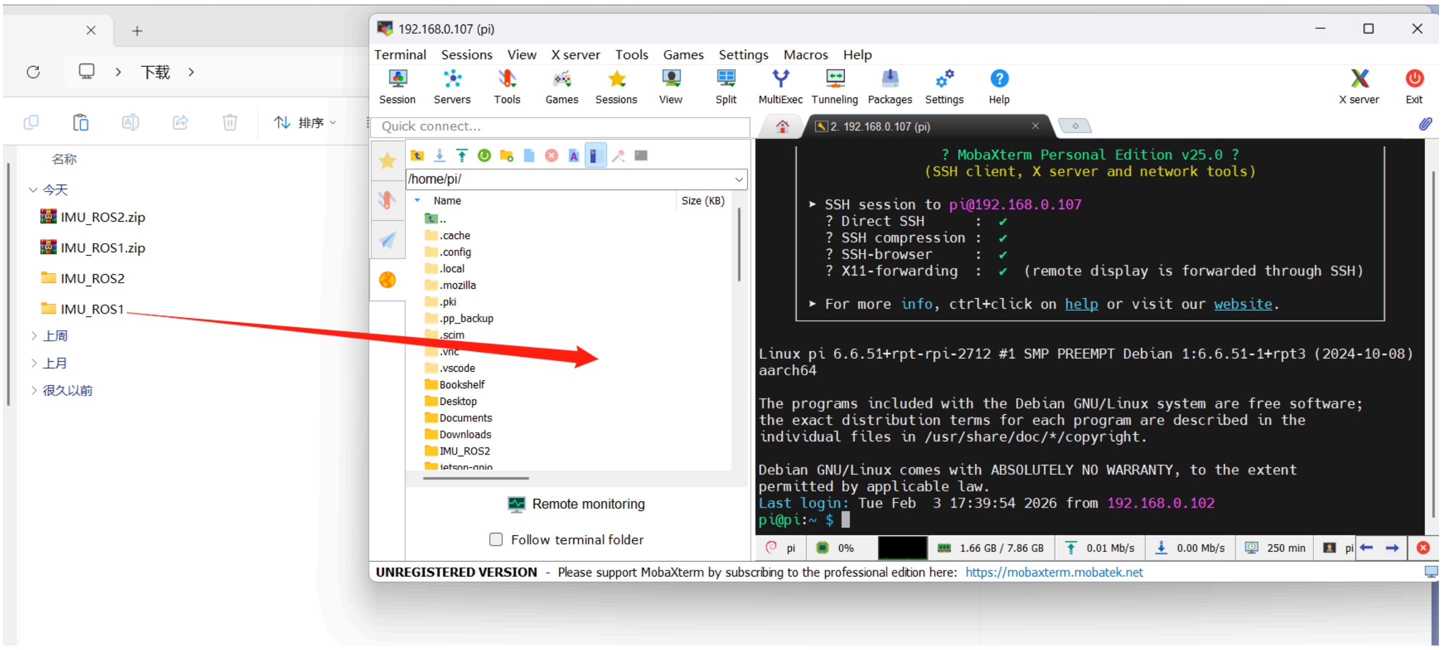
```
sudo apt update
sudo apt install -y python3-serial
sudo apt install -y python3-smbus2
```

3.2 Transfer Files

IMU_ROS2.zip
357.96KB

Friends who are not yet familiar with using MobaXterm to transfer files, please refer to the following webpage for detailed installation and operation methods of MobaXterm: 文件远程传输

Drag the extracted files onto RDK X5 via MobaXterm software.



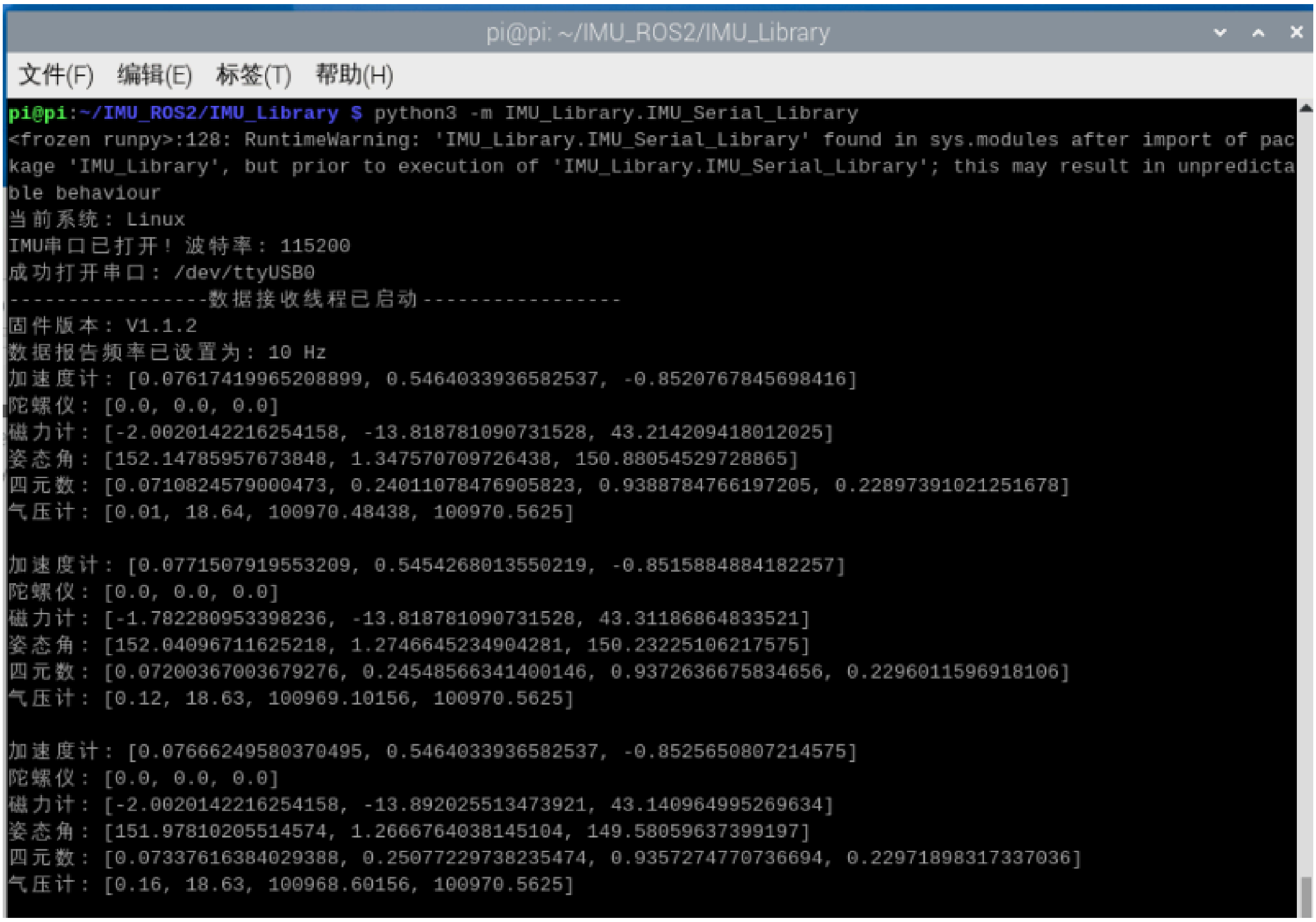
4. View IMU data

Enter the ~/IMU_Library directory and run the IMU_Serial_Library.py file

代码块

```
cd ~/IMU_ROS2/IMU_Library

# 运行 IMU 数据打印文件
python3 -m IMU_Library.IMU_Serial_Library
```




```
pi@pi:~/IMU_ROS2/IMU_Library $ python3 -m IMU_Library.IMU_Serial_Library
<frozen runpy>:128: RuntimeWarning: 'IMU_Library.IMU_Serial_Library' found in sys.modules after import of package 'IMU_Library', but prior to execution of 'IMU_Library.IMU_Serial_Library'; this may result in unpredictable behaviour
当前系统: Linux
IMU串口已打开! 波特率: 115200
成功打开串口: /dev/ttyUSB0
-----------------数据接收线程已启动-----------------
固件版本: V1.1.2
数据报告频率已设置为: 10 Hz
加速度计: [0.07617419965208899, 0.5464033936582537, -0.8520767845698416]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-2.0020142216254158, -13.818781090731528, 43.214209418012025]
姿态角: [152.14785957673848, 1.347570709726438, 150.88054529728865]
四元数: [0.0710824579000473, 0.24011078476905823, 0.9388784766197205, 0.22897391021251678]
气压计: [0.01, 18.64, 100970.48438, 100970.5625]

加速度计: [0.0771507919553209, 0.5454268013550219, -0.8515884884182257]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-1.782280953398236, -13.818781090731528, 43.31186864833521]
姿态角: [152.04096711625218, 1.2746645234904281, 150.23225106217575]
四元数: [0.07200367003679276, 0.24548566341400146, 0.9372636675834656, 0.229601159691810 6]
气压计: [0.12, 18.63, 100969.10156, 100970.5625]

加速度计: [0.07666249580370495, 0.5464033936582537, -0.8525650807214575]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-2.0020142216254158, -13.892025513473921, 43.140964995269634]
姿态角: [151.97810205514574, 1.2666764038145104, 149.58059637399197]
四元数: [0.07337616384029388, 0.25077229738235474, 0.9357274770736694, 0.22971898317337036]
气压计: [0.16, 18.63, 100968.60156, 100970.5625]
```

Note: The above is the data reading for a 10-axis IMU. The 6-axis has no Magnetometer and Barometer data, and the 9-axis has no Barometer data.

5. IMU Calibration

Enter the ~/IMU_Library directory and run the imu_calibration_tool.py file

代码块

```
cd ~/IMU_Library

# 运行 IMU 校准代码文件 --串口通讯校准
# 执行所有校准（整体、磁力计、温度）
python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial

# 仅整体校准
python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial --calibrate imu

# 仅磁力计校准
python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial --calibrate mag

```

```
# 仅温度校准
python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-serial --calibrate temp
```

6. Precautions

If the device ID can be found but the device number cannot, you can refer to the following command to install the ch34x driver

代码块

```
sudo apt remove brltty
git clone https://github.com/clhchan/CH341SER.git
cd CH341SER
make -j6
sudo make install
sudo modprobe ch34x
```